Intel Ethernet Controller E810-CQDA2 - DATASHEET

Product Status: DISCONTINUED

Overview

The Intel Ethernet Controller E810 is a 100 Gigabit Ethernet network controller designed for cloud, enterprise, and telecommunications infrastructure. It supports dual-port QSFP28 100GbE connectivity with full hardware offloading. The E810 has been DISCONTINUED as of August 2025.

Network Specifications

Port speed: 2x 100GbE (QSFP28) or 8x 25GbE breakout. PHY: integrated 100GBASE-CR4/SR4. Ethernet standards: IEEE 802.3by, 802.3bm, 802.3cd. PCIe 4.0 x16 host interface. RSS: 512 queues. VMDq: 256 pools. SR-IOV: up to 256 virtual functions.

Offload Capabilities

RDMA: iWARP and RoCE v2 support. Tunnel offload: VXLAN, GENEVE, GRE, NVGRE. Checksum offload: IPv4, IPv6, TCP, UDP, SCTP. Segmentation offload: TSO, GSO up to 256 KB. RSS with programmable hash. Dynamic Device Personalization (DDP) for custom protocol parsing. Intel Ethernet Flow Director for application-aware traffic steering.

Advanced Features

Precision Time Protocol (PTP/IEEE 1588): hardware timestamping with sub-nanosecond accuracy. Application Device Queues (ADQ) for deterministic latency. Power management: ACPI 2.0 compliant, L0s/L1 ASPM. Management: NC-SI, MCTP sideband. TDP: 18W typical.

Discontinuation Notice

The E810-CQDA2 is discontinued effective August 2025. Last buy date: February 28, 2026. For new designs requiring 100GbE or higher, consider NVIDIA ConnectX-7 (MCX75310AAS-NEAT) offering 400GbE with PCIe Gen5, inline IPsec/TLS encryption, and enhanced RDMA performance. Alternatively, Broadcom BCM57508 provides 200GbE with similar offload capabilities.